Alternating Current Circuits

AC Sources

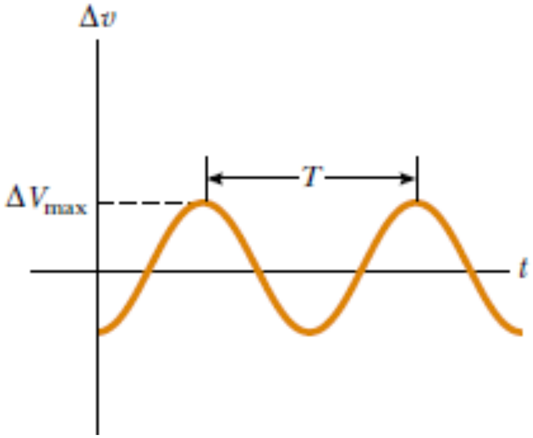
An AC circuit consists of circuit elements and a power source that provides an alternating voltage Δv. This time-varying voltage is described by

$$\Delta v = \Delta V_{\max} \sin \omega t$$

where ΔV_{max} is the maximum output voltage of the AC source, or the **voltage amplitude**.

The angular frequency of the AC voltage is

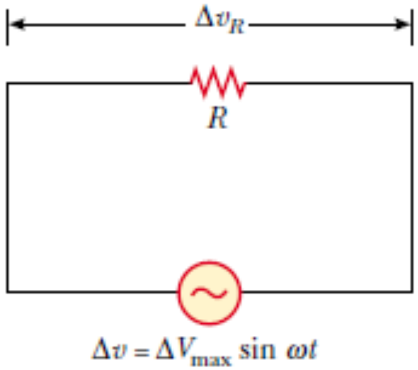
$$\omega = 2\pi f = \frac{2\pi}{T}$$



Resistors in an AC Circuit

that the magnitude of the source voltage equals the magnitude of the voltage across the resistor:

$$\Delta v = \Delta v_R = \Delta V_{\max} \sin \omega t$$



where Δv_R is the **instantaneous voltage across the resistor.** Therefore, from Equation 27.8, $R = \Delta V / I$, the instantaneous current in the resistor is

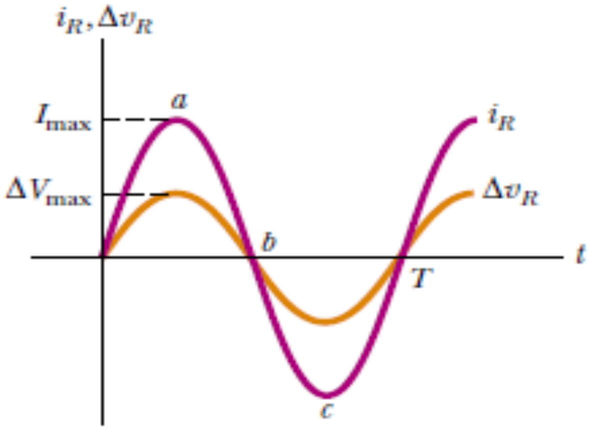
$$i_R = \frac{\Delta v_R}{R} = \frac{\Delta V_{\max}}{R} \sin \omega t = I_{\max} \sin \omega t$$

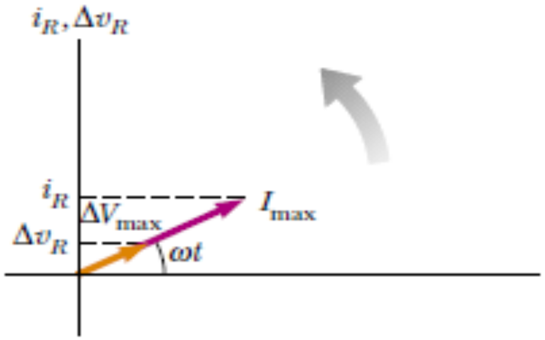
where I_{max} is the maximum current:

$$I_{max} = \frac{\Delta V_{max}}{R}$$

The instantaneous voltage across the resistor is

$$\Delta v_R = I_{max} R \sin \omega t$$

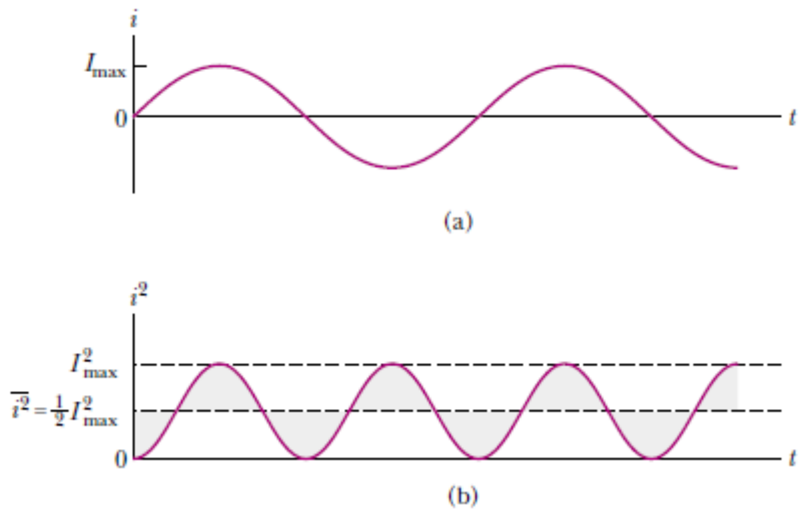




<u>for a sinusoidal applied voltage, the current in a resistor is always in phase with the voltage across the resistor.</u>

The rms current is

$$I_{rms} = \frac{I_{max}}{\sqrt{2}} = 0.707 I_{max}$$



Alternating voltage is also best discussed in terms of rms voltage, and the relationship is identical to that for current:

$$\Delta V_{\mathrm{rms}} = \frac{\Delta V_{\mathrm{max}}}{\sqrt{2}} = 0.707\,\Delta V_{\mathrm{max}}$$

The average power delivered to a resistor that carries an alternating current is

$$\mathcal{P}_{\mathrm{av}} = I_{\mathrm{rms}}^2 R$$

What Is the rms Current?

The voltage output of an AC source is given by the expression $\Delta v = (200\text{ V})\sin\omega t$. Find the rms current in the circuit when this source is connected to a $100\text{-}\Omega$ resistor.

$$\Delta V_{\text{rms}} = \frac{\Delta V_{\text{max}}}{\sqrt{2}} = \frac{200\text{ V}}{\sqrt{2}} = 141\text{ V}$$

$$I_{\text{rms}} = \frac{\Delta V_{\text{rms}}}{R} = \frac{141\text{ V}}{100\ \Omega} = \boxed{1.41\text{ A}}$$

Inductors in an AC Circuit

$$\Delta v - L\frac{di}{dt} = 0$$

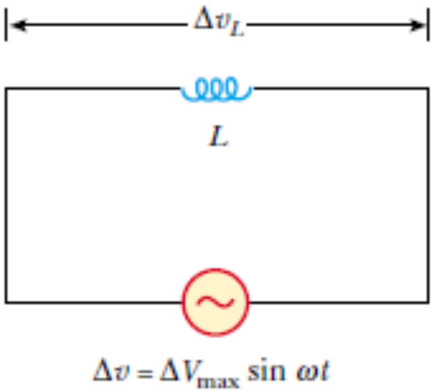
When we substitute $\Delta V_{\max} \sin \omega t$ for Δv and rearrange, we obtain

$$\Delta v = L\frac{di}{dt} = \Delta V_{\max} \sin \omega t$$

we find that

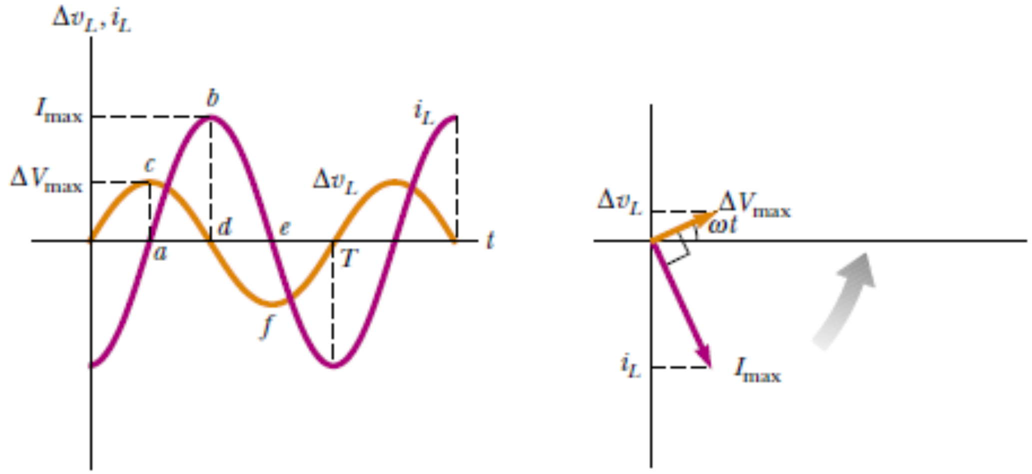
$$di = \frac{\Delta V_{\max}}{L} \sin \omega t \, dt$$

$$i_L = \frac{\Delta V_{\max}}{L}\int \sin \omega t \, dt = -\frac{\Delta V_{\max}}{\omega L} \cos \omega t$$



When we use the trigonometric identity $\cos \omega t = -\sin(\omega t - \pi/2)$

$$i_L = \frac{\Delta V_{\max}}{\omega L} \sin\left(\omega t - \frac{\pi}{2}\right)$$



for a sinusoidal applied voltage, the current in an inductor always lags behind the voltage across the inductor by 90° (one-quarter cycle in time).

$$I_{\max} = \frac{\Delta V_{\max}}{\omega L}$$

we define ωL as the inductive reactance:

$$X_L \equiv \omega L$$

so

$$I_{\max} = \frac{\Delta V_{\max}}{X_L}$$

the instantaneous voltage across the inductor is

$$\Delta v_L = -L\frac{di}{dt} = -\Delta V_{\max} \sin \omega t = -I_{\max} X_L \sin \omega t$$

Example **A Purely Inductive AC Circuit**

In a purely inductive AC circuit, *L = 25.0 mH* and the rms voltage is 150 V. Calculate the inductive reactance and rms current in the circuit if the frequency is 60.0 Hz.

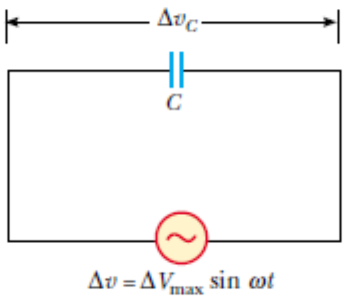
$$X_L = \omega L = 2\pi f L = 2\pi(60.0\text{ Hz})(25.0 \times 10^{-3}\text{ H})$$
$$= \boxed{9.42\ \Omega}$$

$$I_{\text{rms}} = \frac{\Delta V_{L,\text{rms}}}{X_L} = \frac{150\text{ V}}{9.42\ \Omega} = \boxed{15.9\text{ A}}$$

Capacitors in an AC Circuit

The voltage across the capacitor:

$$\Delta v = \Delta v_C = \Delta V_{\max} \sin \omega t$$



$$C = q/\Delta v_C$$

$$q = C\,\Delta V_{\max} \sin \omega t$$

The instantaneous current in the circuit:

$$i_C = \frac{dq}{dt} = \omega C\,\Delta V_{\max} \cos \omega t$$

$$\cos \omega t = \sin\left(\omega t + \frac{\pi}{2}\right)$$

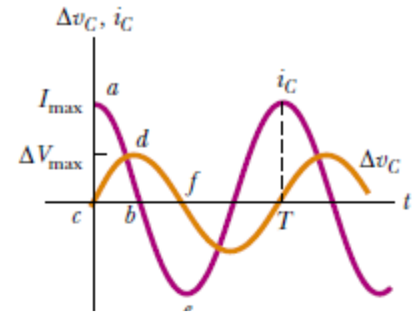
$$i_C = \omega C \Delta V_{\max} \sin\left(\omega t + \frac{\pi}{2}\right)$$

for a sinusoidally applied voltage, the current always leads the voltage across a capacitor by 90°.

The maximum current

$$I_{\max} = \omega C \Delta V_{\max} = \frac{\Delta V_{\max}}{(1/\omega C)}$$

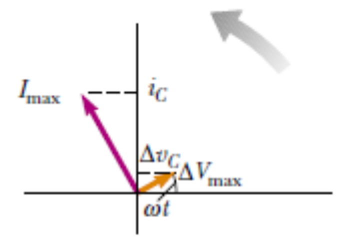
capacitive reactance $$X_C \equiv \frac{1}{\omega C}$$



$$I_{\max} = \frac{\Delta V_{\max}}{X_C}$$

the instantaneous voltage across the capacitor

$$\Delta v_C = \Delta V_{\max} \sin \omega t = I_{\max} X_C \sin \omega t$$



Example **A Purely Capacitive AC Circuit**

An 8.00-μF capacitor is connected to the terminals of a 60.0-Hz AC source whose rms voltage is 150 V. Find the capacitive reactance and the rms current in the circuit.

$\omega = 2\pi f = 377\ \mathrm{s}^{-1}$

$$X_C = \frac{1}{\omega C} = \frac{1}{(377\ \mathrm{s}^{-1})(8.00 \times 10^{-6}\ \mathrm{F})} = \boxed{332\ \Omega}$$

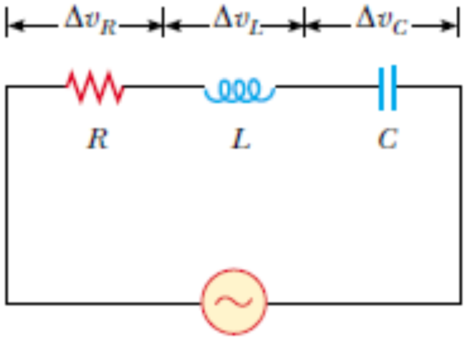
$$I_{\mathrm{rms}} = \frac{\Delta V_{\mathrm{rms}}}{X_C} = \frac{150\ \mathrm{V}}{332\ \Omega} = \boxed{0.452\ \mathrm{A}}$$

The RLC Series Circuit

The instantaneous applied voltage is

$$\Delta v = \Delta V_{\max} \sin \omega t$$

$$i = I_{\max} \sin(\omega t - \phi)$$



the current at all points in a series AC circuit has the same amplitude and phase.

$$\Delta v_R = I_{\max} R \sin \omega t = \Delta V_R \sin \omega t$$

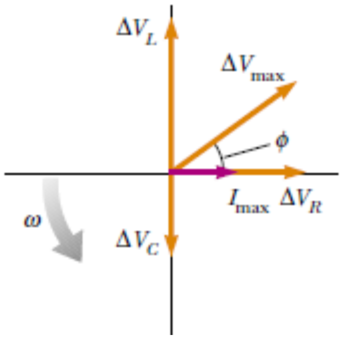
$$\Delta v_L = I_{\max} X_L \sin\left(\omega t + \frac{\pi}{2}\right) = \Delta V_L \cos \omega t$$

$$\Delta v_C = I_{\max} X_C \sin\left(\omega t - \frac{\pi}{2}\right) = -\Delta V_C \cos \omega t$$

The maximum voltage values across the elements:

$$\Delta V_R = I_{\max} R \qquad \Delta V_L = I_{\max} X_L \qquad \Delta V_C = I_{\max} X_C$$

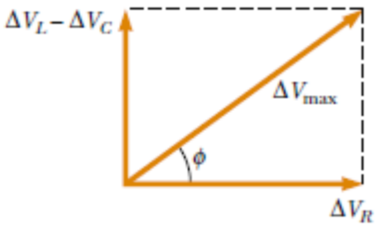
$$\Delta v = \Delta v_R + \Delta v_L + \Delta v_C$$



$$\Delta V_{max} = \sqrt{\Delta V_R^{\,2} + (\Delta V_L - \Delta V_C)^2} = \sqrt{(I_{max}\,R)^2 + (I_{max}\,X_L - I_{max}\,X_C)^2}$$

$$\Delta V_{max} = I_{max}\sqrt{R^2 + (X_L - X_C)^2}$$

$$I_{max} = \frac{\Delta V_{max}}{\sqrt{R^2 + (X_L - X_C)^2}}$$



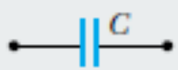
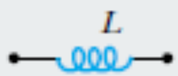
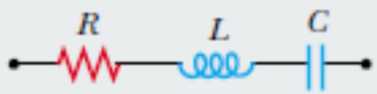
impedance $$Z \equiv \sqrt{R^2 + (X_L - X_C)^2}$$

$$\Delta V_{max} = I_{max}\,Z$$

The phase angle ϕ between the current and the voltage is

$$\phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right)$$

Impedance Values and Phase Angles for Various Circuit-Element Combinations[a]

Circuit Elements	Impedance Z	Phase Angle ϕ
	R	$0°$
	X_C	$-90°$
	X_L	$+90°$
	$\sqrt{R^2 + X_C^2}$	Negative, between $-90°$ and $0°$
	$\sqrt{R^2 + X_L^2}$	Positive, between $0°$ and $90°$
	$\sqrt{R^2 + (X_L - X_C)^2}$	Negative if $X_C > X_L$ Positive if $X_C < X_L$

Example **Analyzing a Series RLC Circuit**

A series *RLC* AC circuit has $R = 425\ \Omega$, $L = 1.25$ H, $C = 3.50\ \mu$F, $\omega = 377\ \text{s}^{-1}$, and $\Delta V_{max} = 150$ V.

(A) Determine the inductive reactance, the capacitive reactance, and the impedance of the circuit.

Solution The reactances are $X_L = \omega L = \boxed{471\ \Omega}$ and $X_C = 1/\omega C = \boxed{758\ \Omega.}$

The impedance is

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$= \sqrt{(425\ \Omega)^2 + (471\ \Omega - 758\ \Omega)^2} = \boxed{513\ \Omega}$$

(B) Find the maximum current in the circuit.

Solution

$$I_{max} = \frac{\Delta V_{max}}{Z} = \frac{150\ \text{V}}{513\ \Omega} = \boxed{0.292\ \text{A}}$$

(C) Find the phase angle between the current and voltage.

Solution

$$\phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right) = \tan^{-1}\left(\frac{471\ \Omega - 758\ \Omega}{425\ \Omega}\right)$$

$$= \boxed{-34.0^\circ}$$

(D) Find both the maximum voltage and the instantaneous voltage across each element.

Solution The maximum voltages are

$$\Delta V_R = I_{\max} R = (0.292\ \text{A})(425\ \Omega) = \boxed{124\ \text{V}}$$

$$\Delta V_L = I_{\max} X_L = (0.292\ \text{A})(471\ \Omega) = \boxed{138\ \text{V}}$$

$$\Delta V_C = I_{\max} X_C = (0.292\ \text{A})(758\ \Omega) = \boxed{221\ \text{V}}$$

The instantaneous voltages across the three elements as

$$\Delta v_R = \boxed{(124\ \text{V}) \sin 377t}$$

$$\Delta v_L = \boxed{(138\ \text{V}) \cos 377t}$$

$$\Delta v_C = \boxed{(-221\ \text{V}) \cos 377t}$$

1- A 10-μF capacitor is plugged into a 110 V-rms 60-Hz voltage source, with an ammeter in series. What is the rms value of the current through the capacitor?

a. 0.202 A (rms)
***b.** 0.415 A* (rms)
c. 0.626 A (rms)
d. 0.838 A (rms)
e. 0.066 A (rms)

2- If an R = 1-kΩ resistor, a C = 1-μF capacitor, and an L = 0.2-H inductor are connected in series with *a* V = 150 sin (377t) volts source, what is the maximum current delivered by the source?

a. 0.007 A
b. 27 mA
***c.** 54 mA*
d. 0.308 A
e. 0.34 A

Power in an AC Circuit

No power losses are associated with pure capacitors and pure inductors in an AC circuit

The instantaneous power P

$$\begin{aligned}\mathcal{P} &= i\,\Delta v = I_{\max}\sin(\omega t - \phi)\,\Delta V_{\max}\sin\omega t \\ &= I_{\max}\,\Delta V_{\max}\sin\omega t\sin(\omega t - \phi)\end{aligned}$$

The average power delivered by the source is converted to internal energy in the resistor

$$\mathcal{P}_{\text{av}} = \tfrac{1}{2} I_{\max}\,\Delta V_{\max}\cos\phi$$

$$\mathcal{P}_{\text{av}} = I_{\text{rms}}\,\Delta V_{\text{rms}}\cos\phi$$

$$\mathcal{P}_{\text{av}} = I_{\text{rms}}^2\,R$$

If $\phi = 0,\ \cos\phi = 1$ $\quad \mathcal{P}_{\text{av}} = I_{\text{rms}}\,\Delta V_{\text{rms}}$

Example **Average Power in an RLC Series Circuit**

Calculate the average power delivered to the series *RLC* circuit described in Example 33.5.

$$\Delta V_{\text{rms}} = \frac{\Delta V_{\max}}{\sqrt{2}} = \frac{150\ \text{V}}{\sqrt{2}} = 106\ \text{V}$$

$$I_{\text{rms}} = \frac{I_{\max}}{\sqrt{2}} = \frac{0.292\ \text{A}}{\sqrt{2}} = 0.206\ \text{A}$$

$$\phi = -34.0°$$

the power factor is $\cos(-34.0°) = 0.829$;

$$\mathcal{P}_{\text{av}} = I_{\text{rms}}\,\Delta V_{\text{rms}} \cos\phi = (0.206\ \text{A})(106\ \text{V})(0.829)$$

$$= \boxed{18.1\ \text{W}}$$

Resonance in a Series RLC Circuit

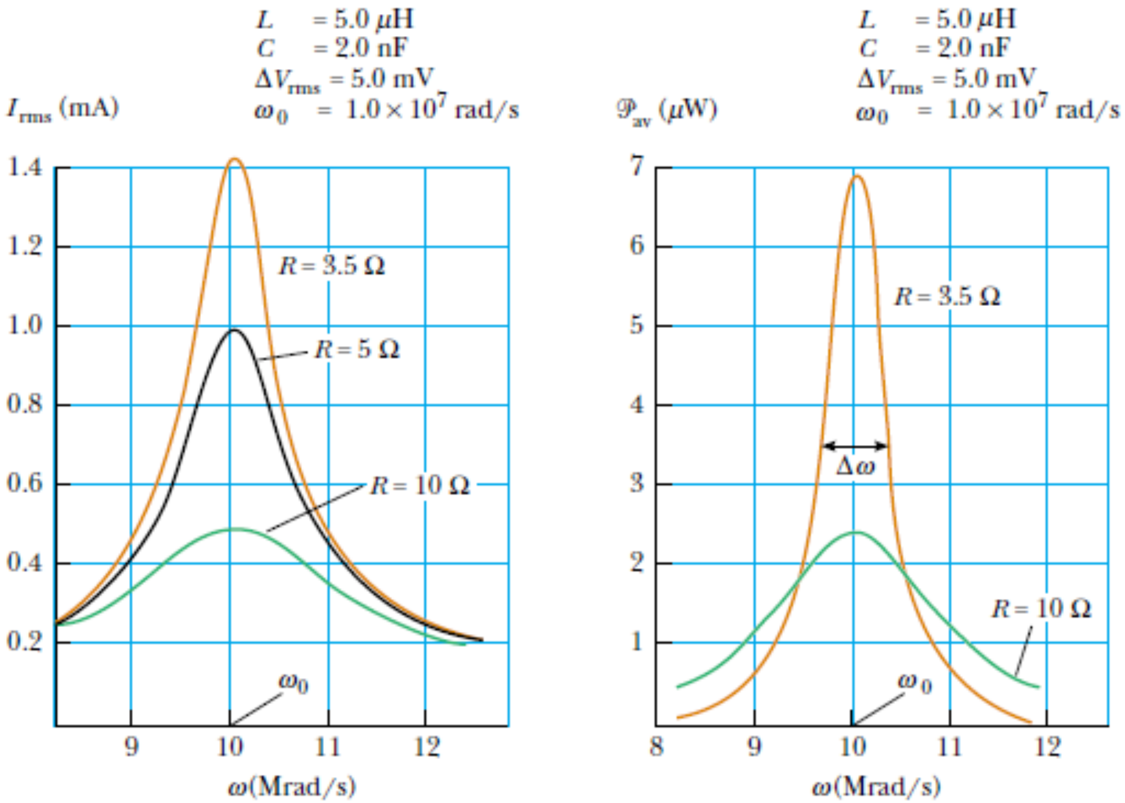
A series *RLC* circuit is said to be **in resonance** when the current has its maximum value. In general, the rms current can be written

$$I_{rms} = \frac{\Delta V_{rms}}{Z}$$

Because the impedance depends on the frequency of the source, the current in the *RLC* circuit also depends on the frequency. The frequency ω_0 at which $X_L - X_C = 0$ is called the **resonance frequency** of the circuit. To find ω_0, we use the condition $X_L = X_C$, from which we obtain $\omega_0 L = 1/\omega_0 C$, or

$$\boxed{\omega_0 = \frac{1}{\sqrt{LC}}}$$

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{LC}}$$



$L = 5.0\ \mu H$
$C = 2.0\ nF$
$\Delta V_{rms} = 5.0\ mV$
$\omega_0 = 1.0 \times 10^7\ rad/s$
I_{rms} (mA)
1.4
1.2
1.0
0.8
0.6
0.4
0.2
$R = 3.5\ \Omega$
$R = 5\ \Omega$
$R = 10\ \Omega$
ω_0
9
10
11
12
ω(Mrad/s)
$L = 5.0\ \mu H$
$C = 2.0\ nF$
$\Delta V_{rms} = 5.0\ mV$
$\omega_0 = 1.0 \times 10^7\ rad/s$
$\mathcal{P}_{av}$ (μW)
7
6
5
4
3
2
1
$R = 3.5\ \Omega$
$\Delta\omega$
$R = 10\ \Omega$
ω_0
8
9
10
11
12
ω(Mrad/s)

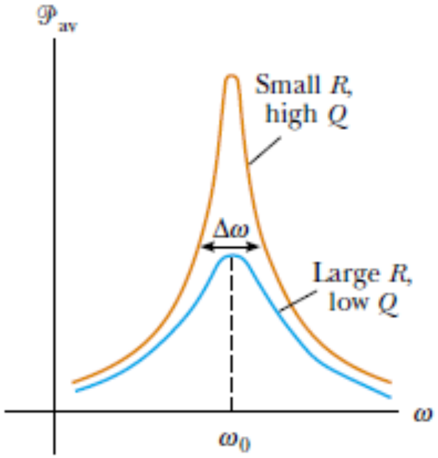
versus frequency for two values of R in a series RLC circuit. As the resistance is made smaller, the curve becomes sharper in the vicinity of the resonance frequency. This curve sharpness is usually described by a dimensionless parameter known as the **quality factor,**[3] denoted by Q:

$$Q = \frac{\omega_0}{\Delta\omega}$$

$$\Delta\omega = R/L,$$

Quality factor

$$Q = \frac{\omega_0 L}{R}$$



Example **A Resonating Series RLC Circuit**

Consider a series RLC circuit for which $R = 150\ \Omega$, $L = 20.0$ mH, $\Delta V_{\text{rms}} = 20.0$ V, and $\omega = 5\,000\ \text{s}^{-1}$. Determine the value of the capacitance for which the current is a maximum.

$$\omega_0 = 5.00 \times 10^3\ \text{s}^{-1} = \frac{1}{\sqrt{LC}}$$

$$C = \frac{1}{\omega_0{}^2 L} = \frac{1}{(5.00 \times 10^3\ \text{s}^{-1})^2 (20.0 \times 10^{-3}\ \text{H})}$$

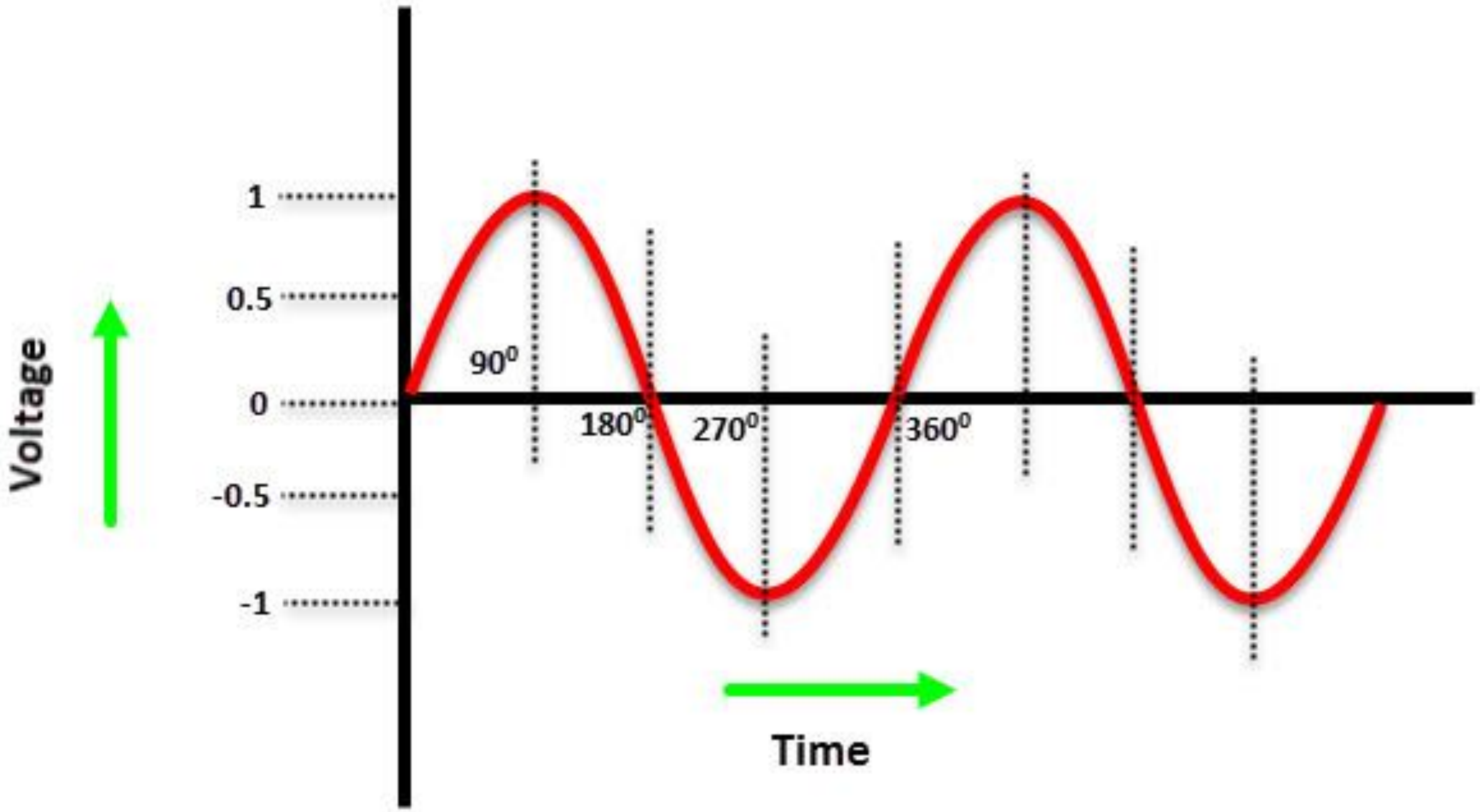
$$= \boxed{2.00\ \mu\text{F}}$$

3 Phase System

Difference Between Single Phase and Three Phase

A Single Phase AC Power system consists of two wires known as the phase (or Line or Live or Hot) and the neutral wire.

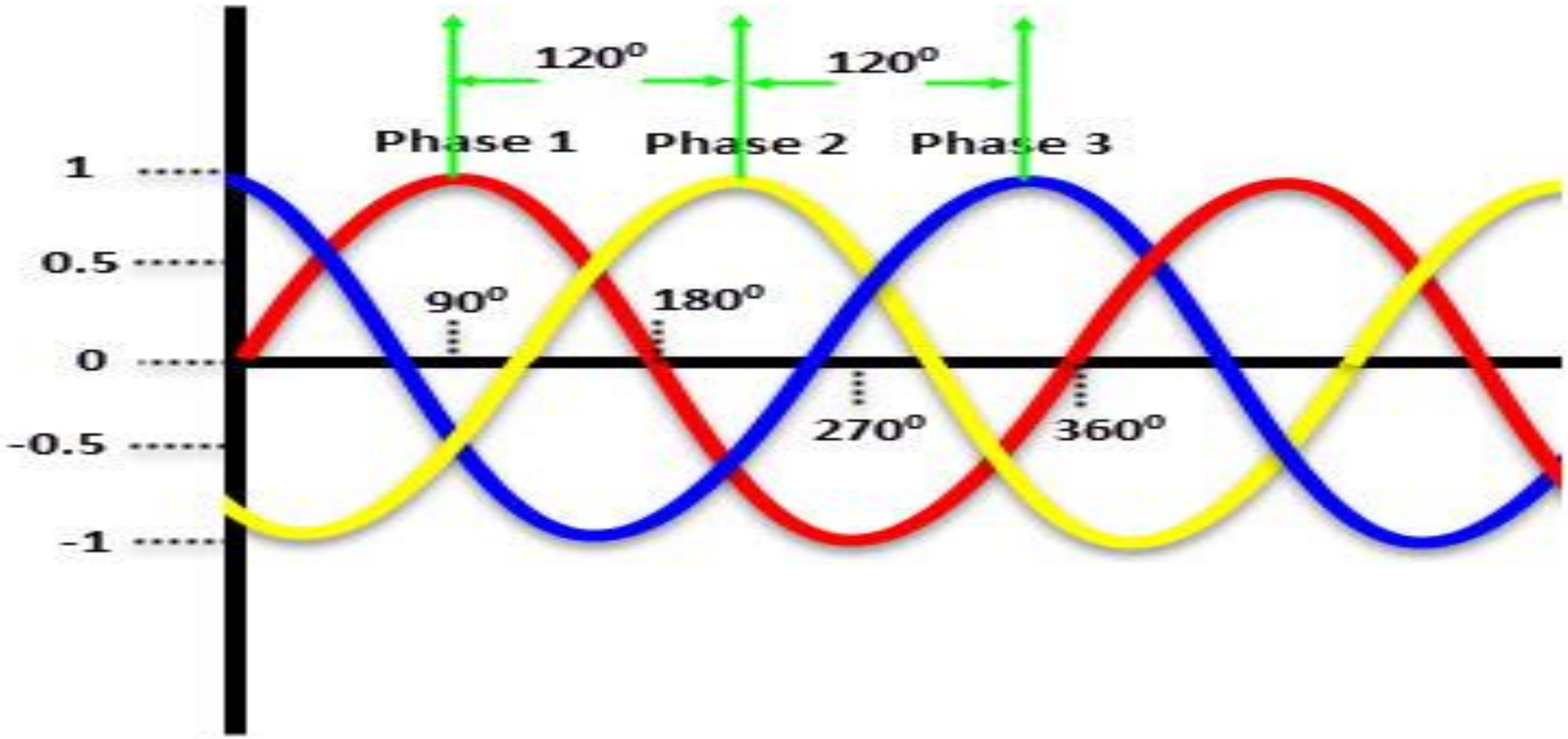
In three phase system, there are either three wires or four wires for transmitting power (no neutral in three wire three phase power and all the three wires are phases).



Single Phase

Usually, the single phase voltage is 230V and the frequency is 50Hz

Since the voltage in a single phase supply rises and falls (peaks and dips), a constant power cannot be delivered to the load.



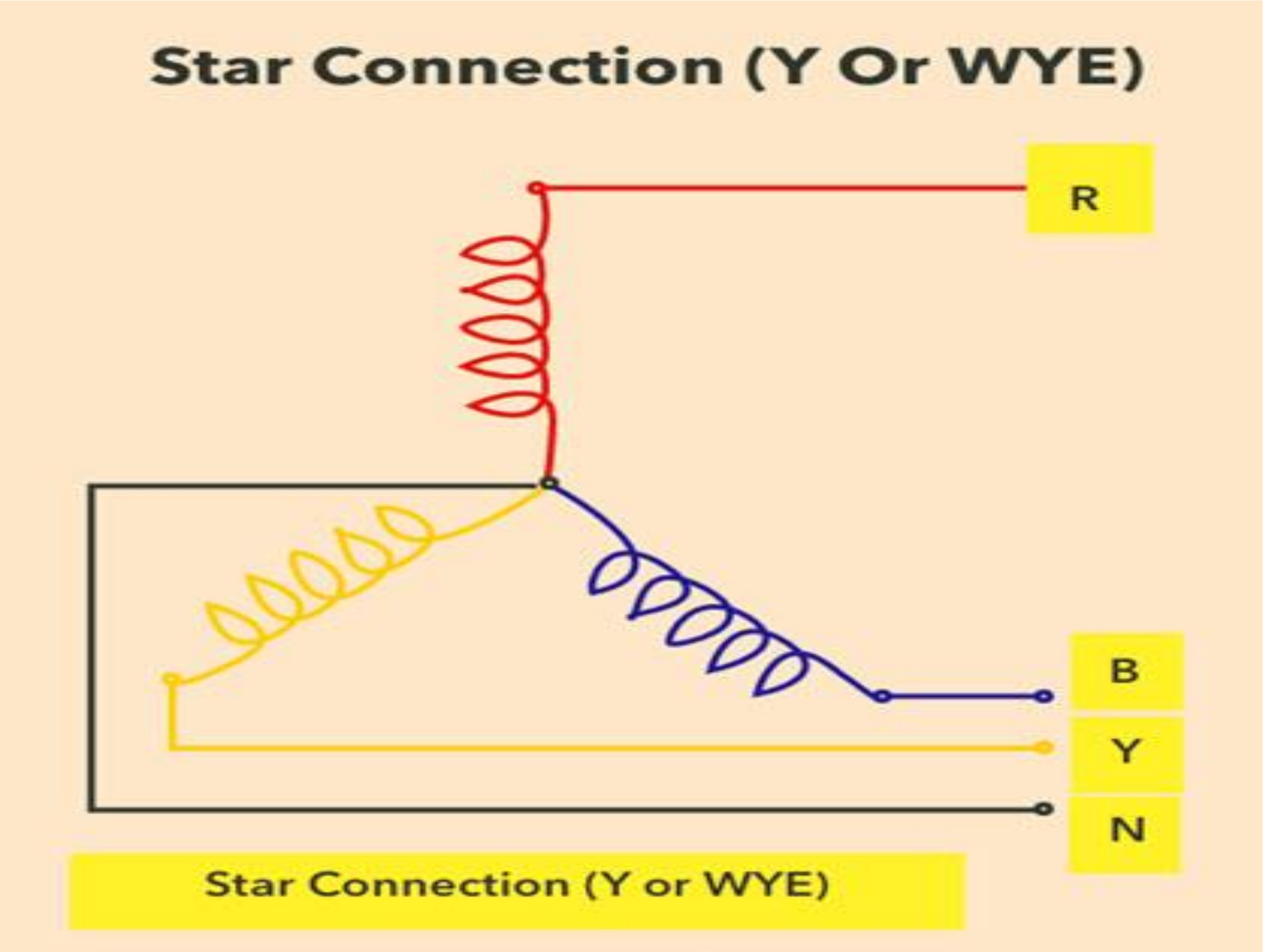
3 Phase supply

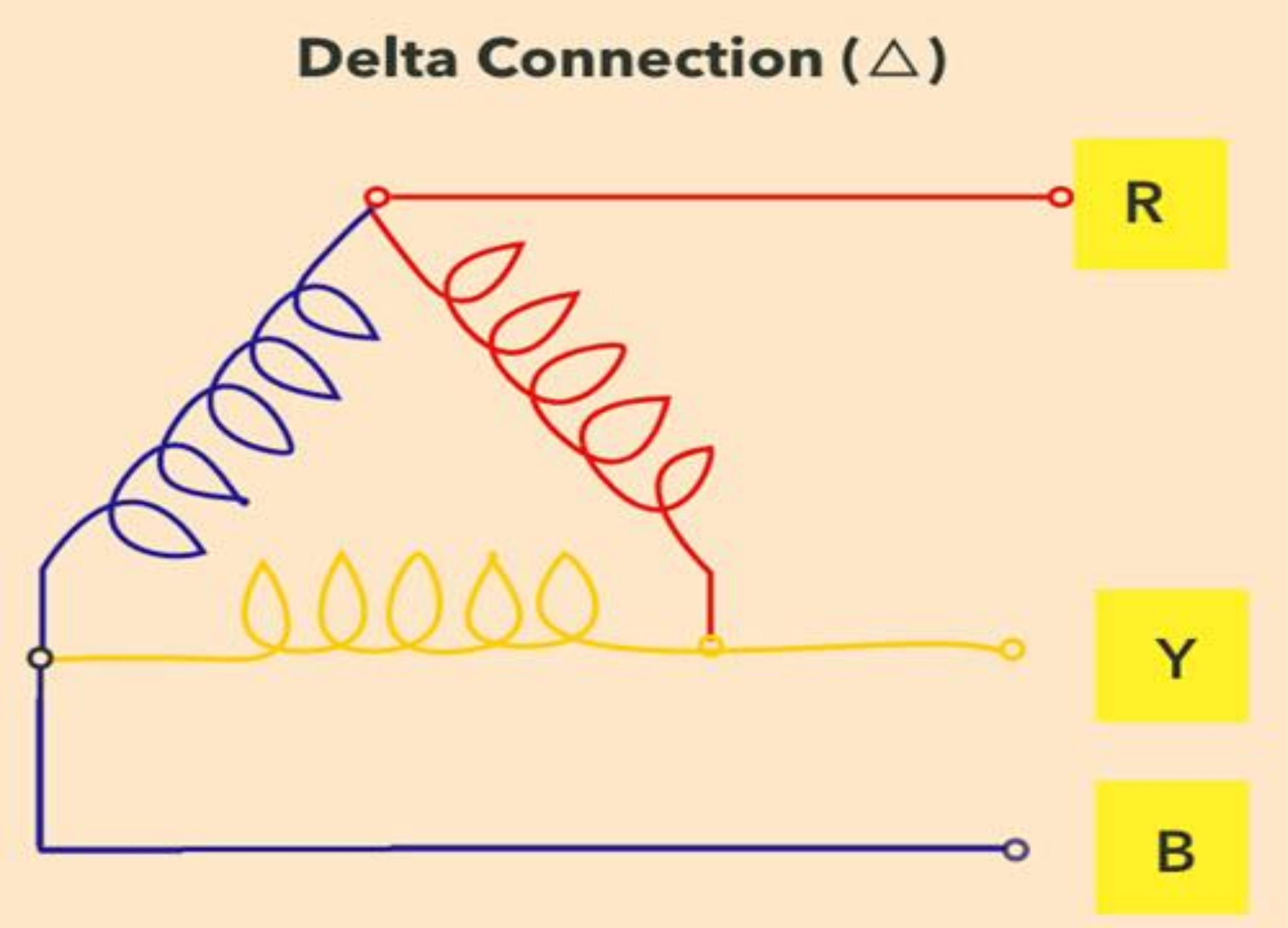
In a three phase power supply, the power never drops to zero.

This steady stream of power and ability to handle higher loads makes a three phase supply suitable for industrial and commercial operations.

There are two types of circuit configurations in a three phase power supply. They are the Delta and the Star (Y or Wye).

The voltage of single phase supply is 230V whereas it is 415V in a three phase supply.


Star Connection (Y Or WYE)
R
B
Y
N
Star Connection (Y or WYE)


Delta Connection (△)
R
Y
B

Advantages

- For the same power, a three phase power supply uses less wires.
- Three phase power supply is usually the preferred network for commercial and industrial loads.
- Studies found that a three phase power supply is more economical and easy to produce.
- The overall efficiency of the three phase power supply is higher when compared to that of a single phase power supply for the same load.

Terms used in three-phase systems and circuits

(i) Balanced supply: a set of three sinusoidal voltages (or currents) that are equal in magnitude but has a phase difference of 120º, constitute a balanced three-phase voltage (or current) system.

(ii) Unbalanced supply: a three-phase system is said to be unbalanced when either of the three-phase voltages are unequal in magnitude or the phase angle between the three phases is not equal to 120º.

(iii) Balanced load: if the load impedances of the three phases are identical in magnitude as well as phase angle, then the load is said to be balanced. It implies that the load has the same value of resistance R and reactance XL and/or XC in each phase.

(iv) Unbalanced load: if the load impedances of the three phases are neither identical in magnitude nor in phase angle then the load is said to be unbalanced.

(v) Single phasing: when one phase of the three-phase supply is not available then the condition is called single phasing.

(vi) Phase sequence: the order in which the maximum value of voltages of each phase appear is called the phase sequence. It can be RYB or RBY.

(vii) Coil: a coil is made of conducting wire, say copper, having an insulation cover. A coil can be of a single turn or many number of turns. Normally a coil will have a number of turns. A single turn of a coil will have two conductors on its two sides called coil sides

(viii) Winding: a number of coils are used to make one winding. Normally the winding coils are connected in series. One winding forms one phase.

(ix) Symmetrical system: in a symmetrical three-phase system the magnitude of three-phase voltages is the same but there is a time phase difference of 120 deg between the voltages.

Three-Phase Winding Connections

A three-phase generator will have three-phase windings. These phase windings can be connected in two ways:

1. Star connection

2. Delta connection

Star Connection

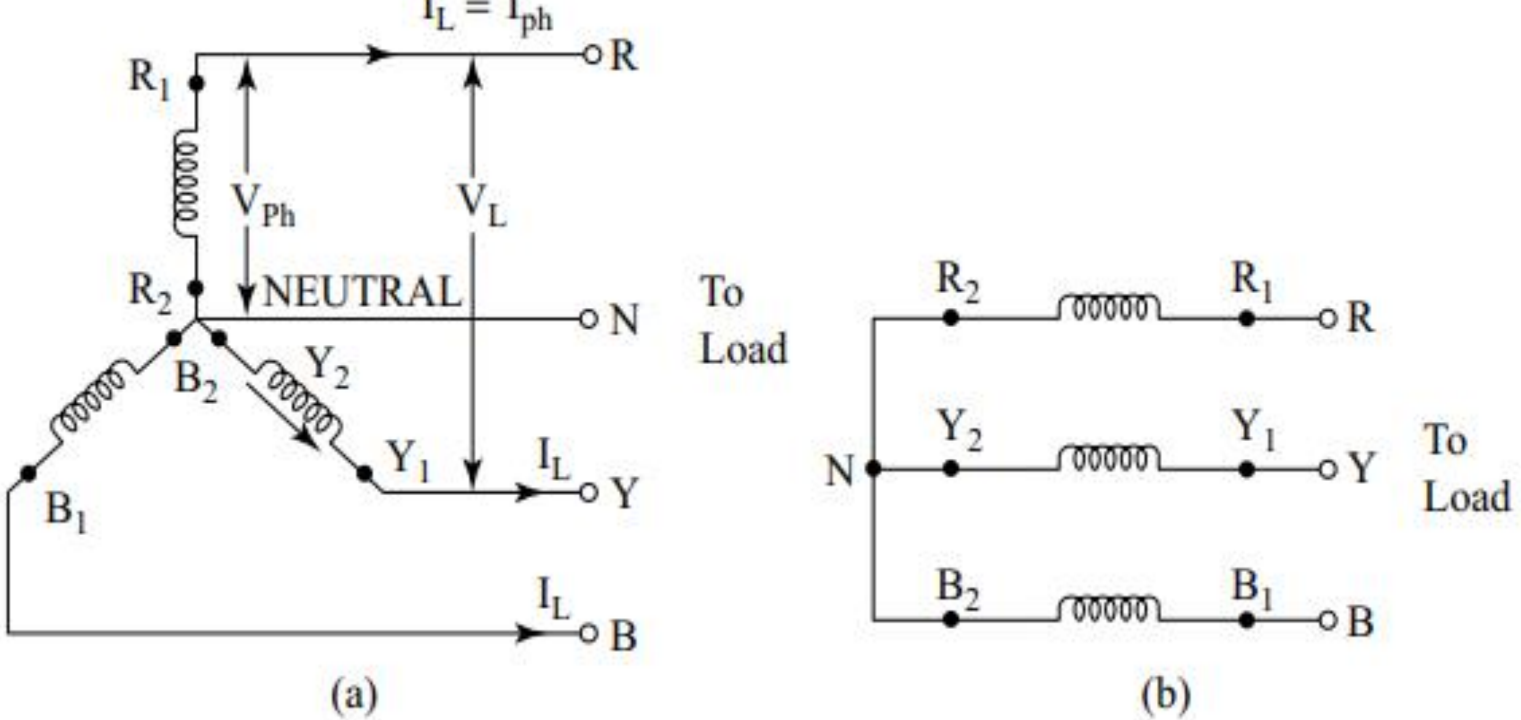


Figure 4.6 *Star connection of phase windings: (a) three-phase four-wire system; (b) three-phase three-wire system*

The star connection is shown in Fig. 4.6 (a). This is a three-phase, four-wire star-connected system. If no neutral conductor is taken out from the system it gives rise to a three-phase, three-wire star-connected system.

- The current flowing through each line conductor is called line current, I_L. In the star connection the line current is also the phase current. Similarly, voltage across each phase is called phase voltage, V_{Ph}.

- Voltage across any two line conductors is called line voltage, V_L. When a balanced three-phase load is connected across the supply terminals R, Y, B currents will flow through the circuit.

- The sum of these currents, i.e., I_R, I_Y, and I_B will be zero. The neutral wire connected between the supply neutral point and the load neutral point will carry no current for a balanced system.

Delta Connection

The delta connection is formed by connecting the end of one winding to the starting end of the other and connections are continued to form a closed loop.

- In this case, the current flowing through each line conductor is called line current I_L and the current flowing through each phase winding is called phase current, I_{Ph}.
- However, we find that the phase voltage is the same as the line voltage in a delta connection.

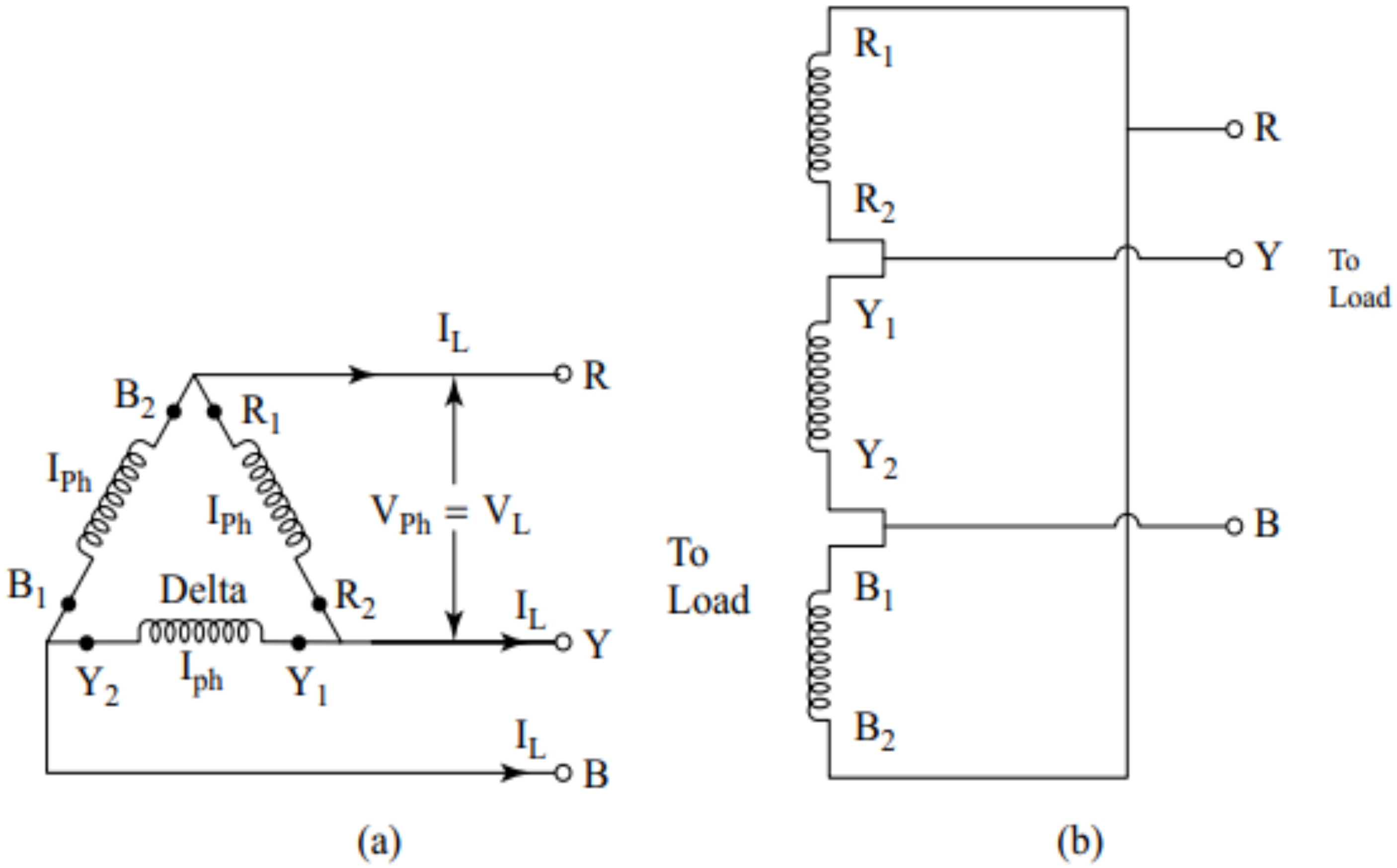


Figure 4.7 *Delta connection of three-phase windings: (a) three-phase three-wire delta connection; (b) connection scheme of windings forming a delta*

Relationship of Line and Phase Voltages, and Currents in a Star-connected System

Suppose load is inductive and, therefore, current will lag the applied voltage by angle phi

Consider a balanced system so that the magnitude of current and voltage of each phase will be the same.

i.e., Phase voltages,	$V_R = V_Y = V_B = V_{Ph}$
Line current,	$I_R = I_Y = I_B = I_L$
Line voltage,	$V_L = V_{RY} = V_{YB} = V_{BR}$
Phase current,	$I_{Ph} = I_R = I_Y = I_B$

$I_L = I_{PH}$ for star connection as the same phase current passes through the lines to the load.

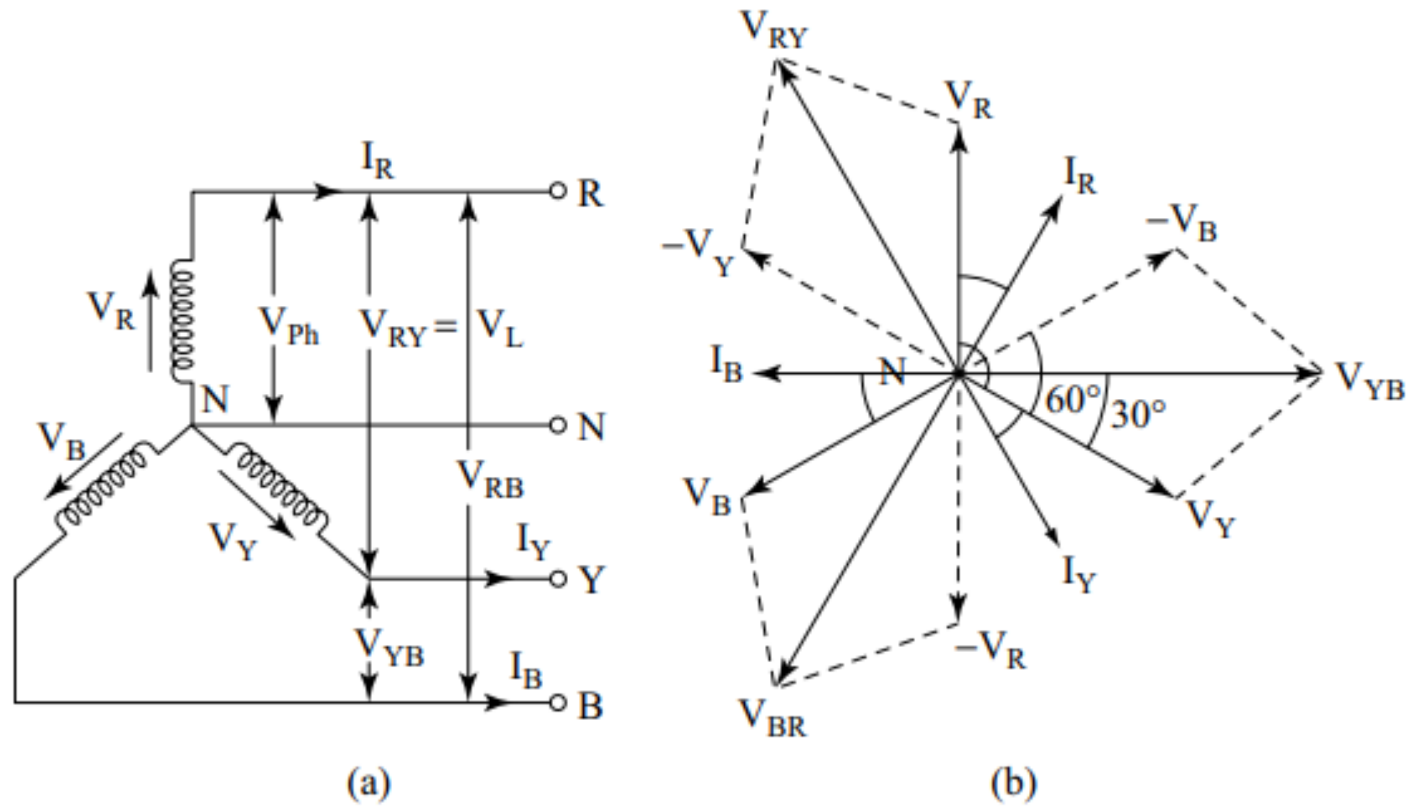


Figure 4.8 *(a) Balanced star-connected system; (b) phasor diagram*

To derive the relation between V_L and V_{Ph}, consider line voltage V_{RY}.

$$V_{RY} = V_{RN} + V_{NY}$$

$$V_{RY} = V_{RN} + (-V_{YN})$$

Similarly

$$V_{YB} = V_{YN} + (-V_{BN})$$

and

$$V_{BR} = V_{BN} + (-V_{RN})$$

$$V_{RY} = \sqrt{V_R{}^2 + V_Y{}^2 + 2V_R V_Y \cos 60°}$$

$$V_{RY} = V_L = \sqrt{V_{Ph}{}^2 + V_{Ph}{}^2 + 2V_{Ph} V_{Ph} \times \frac{1}{2}}$$

$$V_L = \sqrt{3}\, V_{Ph}$$

$$V_L = \sqrt{3}\, V_{Ph}$$

Power: Power output per phase $= V_{Ph} I_{Ph} \cos\phi$

Total power output $= 3 V_{Ph} I_{Ph} \cos\phi$

$$= 3 \times \frac{V_L}{\sqrt{3}} \times I_L \cos\phi$$

$$P = \sqrt{3}\, V_L\, I_L \cos\phi$$

Power = $\sqrt{3}$ × line voltage × line current × power factor

Relationship of Line and Phase Voltages and Currents in a Delta-connected System

In a delta-connected system the voltage across the winding, i.e., the phases is the same as that across the line terminals. However, the current through the phases is not the same as through the supply lines.

Therefore, in the case of the delta-connected circuit, phase voltage is equal to the line voltage, but line current is not equal to phase current.

Line voltage	$V_L = V_{RY} = V_{YB} = V_{BR}$
Line current	$I_L = I_R = I_Y = I_B$
Phase voltage	$V_{Ph} = V_{RY} = V_{YB} = V_{BR}$
Phase current	$I_{Ph} = I_{RY} = I_{YB} = I_{BR}$

$V_{Ph} = V_L$ for delta-connected load

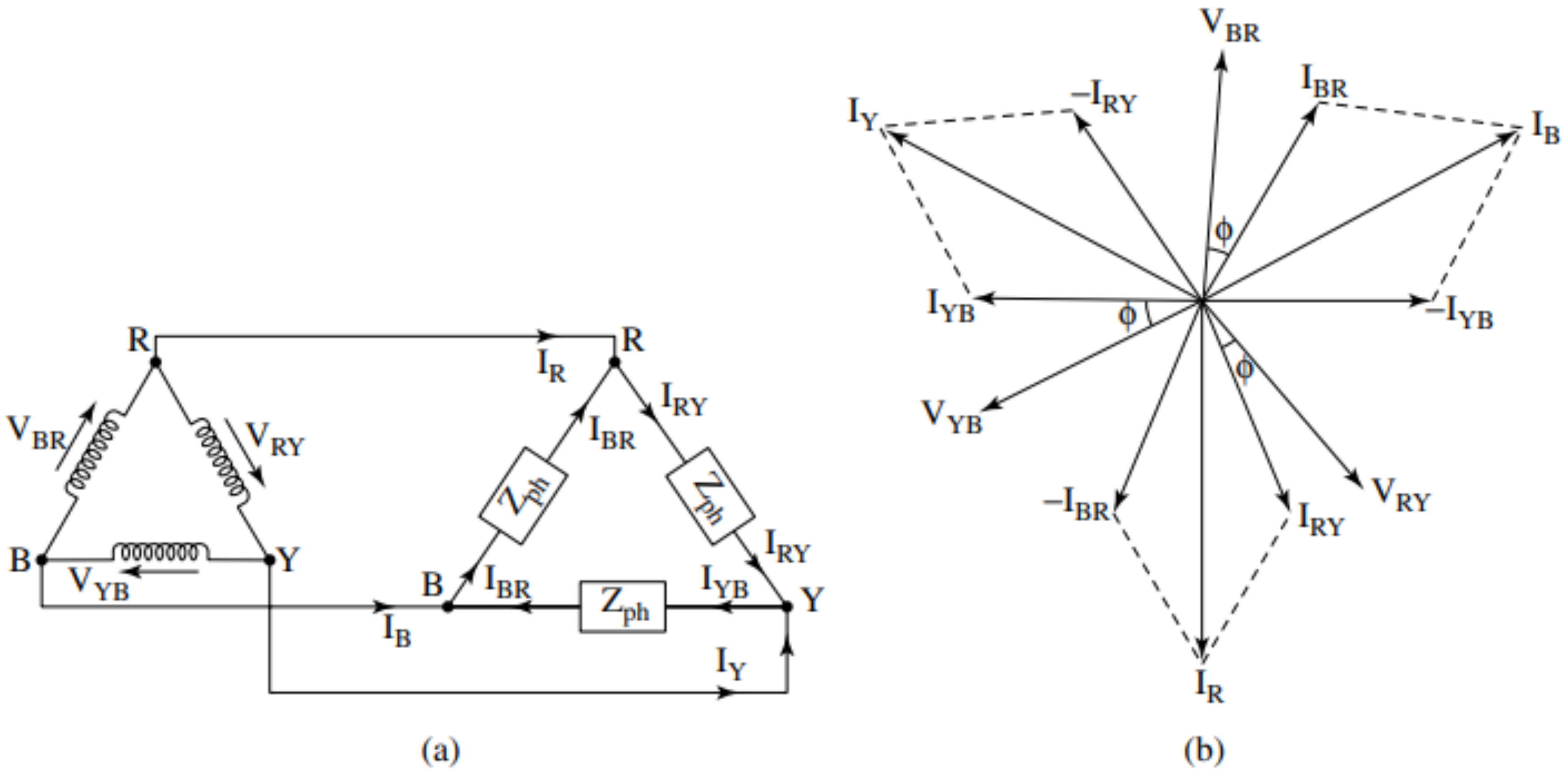


Figure 4.9 *Delta-connected system: (a) a three-phase delta-connected load supplied from a delta-connected supply source; (b) phasor diagram of voltages and currents*

- The line currents are I_R, I_Y, and I_B, respectively. The phase currents are I_{RY}, I_{YB}, and I_{BR}.
- The phasor diagram in Fig. 4.9 (b) has been developed by first showing the three-phase voltages V_{YB}, V_{BR}, and V_{RY} of equal magnitude but displaced by 120° from each other.
- Then the phase currents I_{YB}, I_{BR}, and I_{RY} have been shown lagging respective phase voltages by power factor angle phi.
- The line currents are drawn by applying KCL at the nodes R, Y, and B and adding the phasors, as has been shown.

Apply KCL at node R

$$I_R + I_{BR} = I_{RY}$$

$$\therefore \quad I_R = I_{RY} - I_{BR}$$

Similarly at node Y and B, we can write

$$I_Y = I_{YB} - I_{RY}$$

$$I_B = I_{BR} - I_{YB}$$

Since phase angle between phase currents I_{RY} and $-I_{BR}$ is 60°

$\therefore$

$$I_R = \sqrt{I_{RY}^2 + I_{BR}^2 + 2I_{RY} I_{BR} \cos 60^\circ}$$

$$I_R = I_L = \sqrt{I_{Ph}^2 + I_{Ph}^2 + 2I_{Ph} I_{Ph} \times \frac{1}{2}}$$

$$I_L = \sqrt{3} \times I_{ph}$$

Thus, for a three-phase delta-connected system,

Line current = $\sqrt{3}$ × Phase current

Line voltage = Phase voltage

Power: Power output per phase $= V_{Ph} I_{Ph} \cos \phi$

Total power output $= 3V_{Ph} I_{Ph} \cos \phi$

$$= 3 \times V_L \times \frac{I_L}{\sqrt{3}} \cos \phi$$

$$\sqrt{3} V_L I_L \cos \phi$$

$$\text{Power} = \sqrt{3} \times \text{Line voltage} \times \text{Line current} \times \text{Power factor}$$

For both star-connected and delta-connected systems, the total power P is

$$P = \sqrt{3}\, V_L\, I_L \cos\phi$$

If per phase power is P_h and total power is P_T,

then

$$P_T = 3\, P_h$$

Comparison between Star connection and Delta connection

Star Connection	Delta Connection
1. Line current is the same as phase current, i.e., $I_L = I_{Ph}$	1. Line current is $\sqrt{3}$ $\times$ the phase current, i.e., $I_L = \sqrt{3}\, I_{Ph}$
2. Line voltage is $\sqrt{3}$ the phase voltage, i.e., $V_L = \sqrt{3}\, V_{Ph}$	2. Line voltage is the same as phase voltage, i.e., $V_L = V_{ph}$
3. Total power = $\sqrt{3}\, V_L\, I_L \cos\phi$	3. Total power = $\sqrt{3}\, V_L\, I_L \cos\phi$
4. Per phase power = $V_{Ph}\, I_{Ph} \cos\phi$	4. Per phase power = $V_{Ph}\, I_{Ph} \cos\phi$
5. Three-phase three-wire and three-phase four-wire systems are possible	5. Three-phase three-wire system is possible
6. Line voltages lead the respective phase voltages by 30°	6. Line currents lag the respective phase currents by 30°

Measurement of power in three-phase circuits

In three-phase ac circuits, total power is three times the power per phase.

Wattmeter is an instrument used for measurement of power in ac circuits. Wattmeters are available as single-phase wattmeters and three-phase wattmeters.

One-Wattmeter Method

In this method, only one single-phase wattmeter can be used to measure the total three-phase power. In this method, the current coil (CC) of the wattmeter is connected in series with any phase and the pressure coil (PC) is connected between that phase and the neutral as shown in Fig. 4.15.

In this method one wattmeter will measure only the power of one phase. Hence, total power is taken as three times the wattmeter reading.

$$\text{Total Power} = 3 \times V_{ph} I_{ph} \cos \phi$$

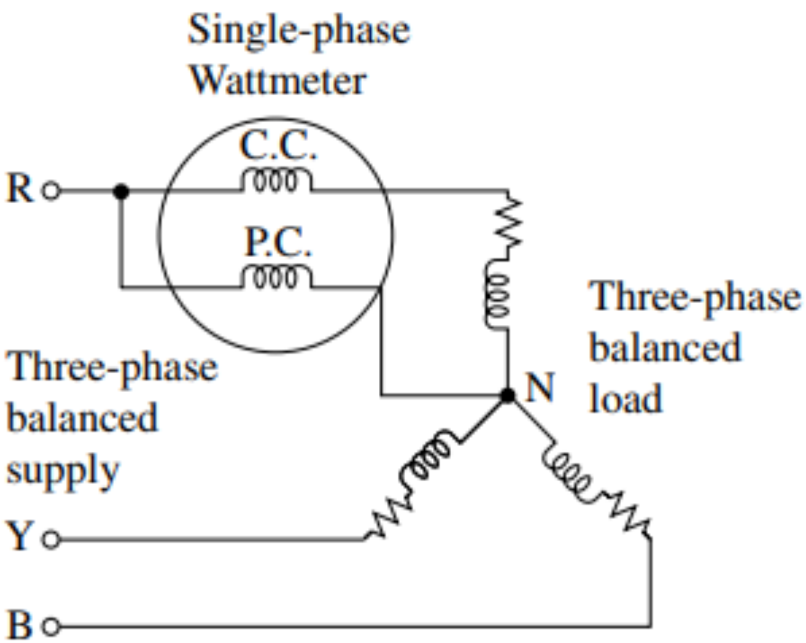


Figure 4.15 *One-wattmeter method of measuring power of a star-connected balanced three-phase load*

Two-Wattmeter Method

This method requires only two wattmeters to measure three-phase power for balanced as well as unbalanced loads.

In this method two wattmeters are connected in two phases and their pressure coils are connected to the remaining third phase as has been shown in Fig. 4.16.

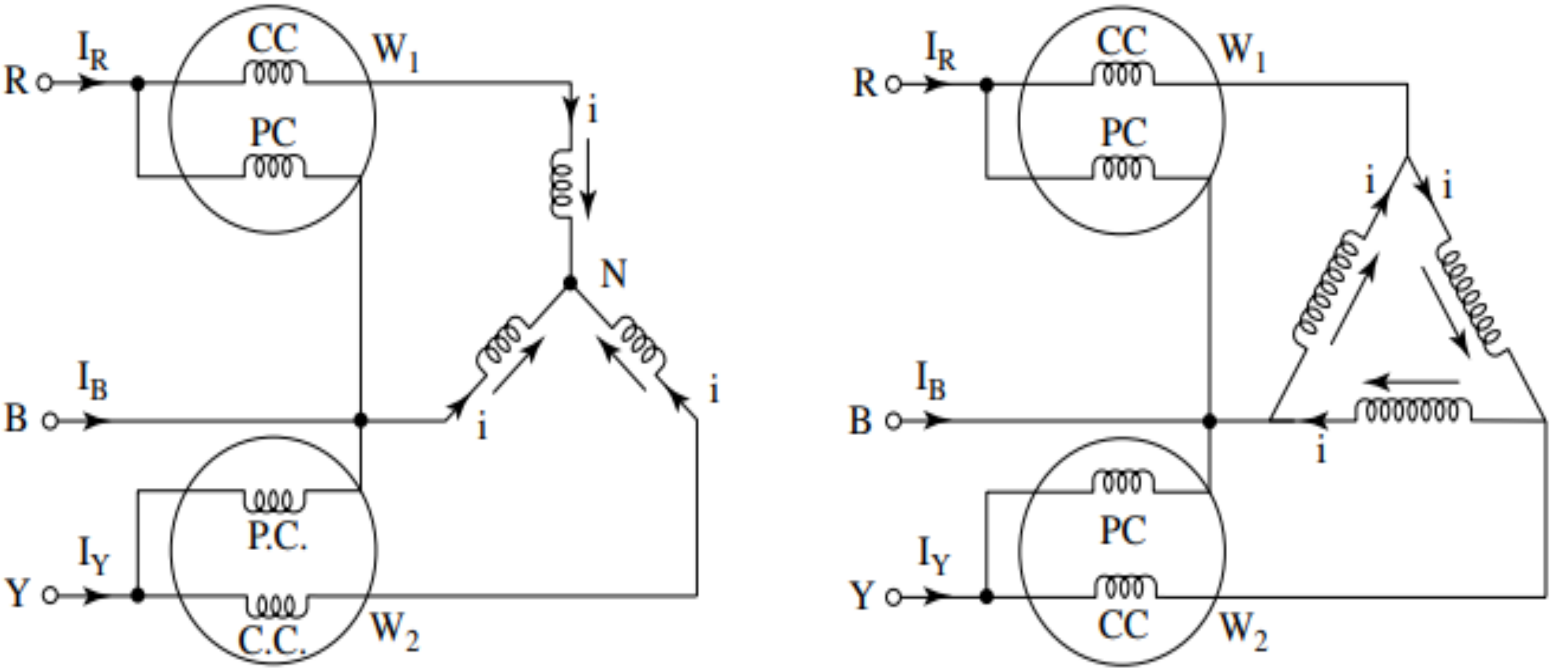


Figure 4.16 *Two-wattmeter method of measuring power for star- and delta-connected load*

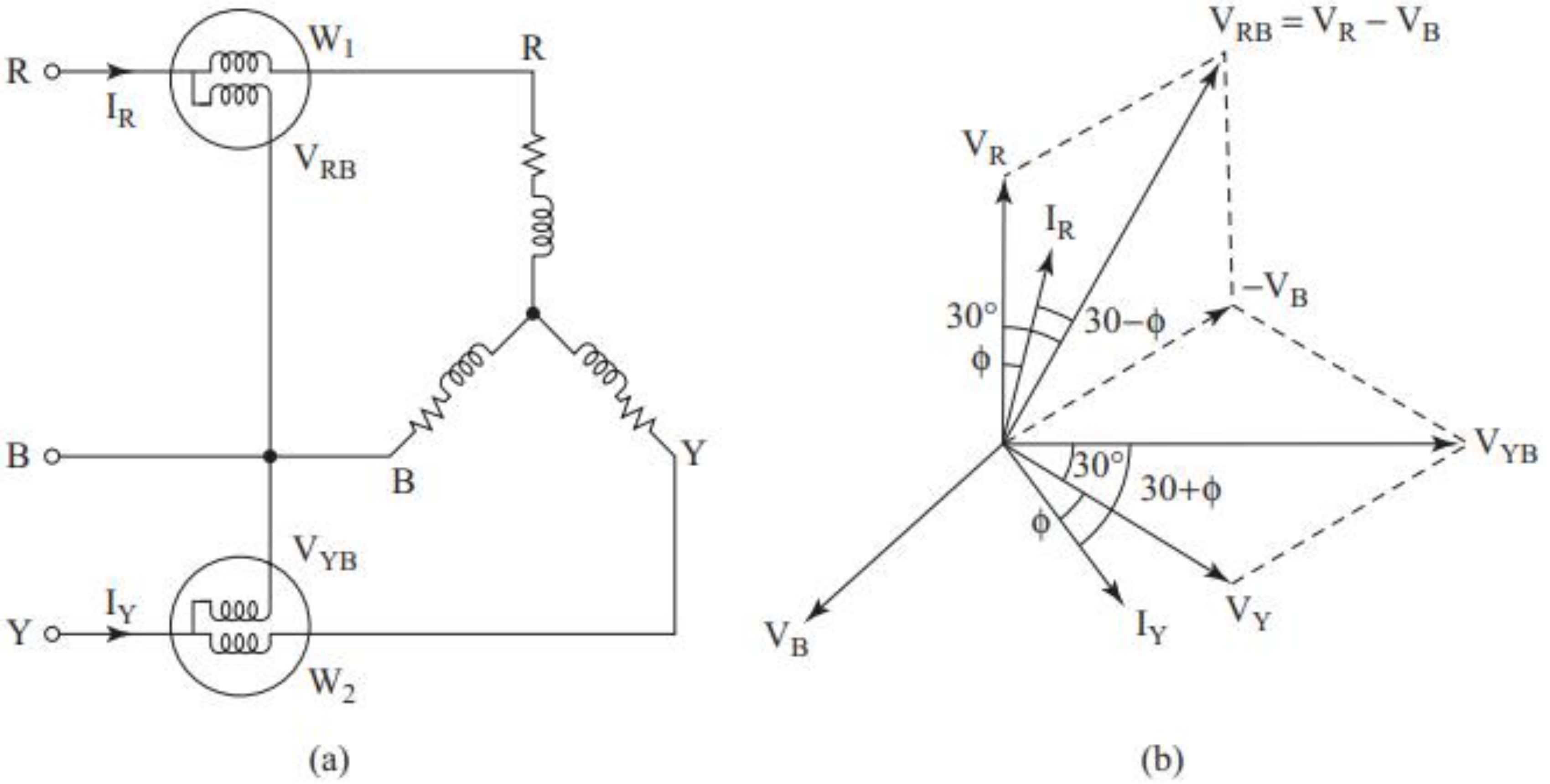


Figure 4.17 *(a) Measurement of three-phase power using two single-phase wattmeters; (b) phasor diagram*

- Let W_1 and W_2 respectively be the two wattmeter readings.
- Current flowing through the current coil of wattmeter W_1 is I_R.
- The voltage appearing across its pressure coil is V_{RB}.
- The wattmeter reading will be equal to, $W_1 = V_{RB} I_R$. cos of angle between V_{RB} and I_R.
- Similarly, the wattmeter reading W_2 will be equal to, $W_2 = V_{YB} I_B$ cos of angle between V_{YB} and I_B.

From the phasor diagram as shown in Fig. 4.17 (b),

$$W_1 = V_{RB}\, I_L \cos(30-\phi) = \sqrt{3}\, V_{ph}\, I_{ph} \cos(30-\phi) = V_L I_L \cos(30-\phi)$$

And

$$W_2 = V_{YB}\, I_Y \cos(30+\phi) = \sqrt{3}\, V_{ph}\, I_{ph} \cos(30+\phi) = V_L I_L \cos(30+\phi)$$

Let us add the two wattmeter readings, i.e., W_1 and W_2.

$$\begin{aligned} W_1 + W_2 &= \sqrt{3}\, V_{ph}\, I_{ph} \cos(30-\phi) + \sqrt{3}\, V_{ph}\, I_{ph} \cos(30+\phi) \\ &= \sqrt{3}\, V_{ph}\, I_{ph} [\cos(30-\phi) + \cos(30+\phi)] \\ &= \sqrt{3}\, V_{ph}\, I_{ph}\, 2\cos\phi \cos 30^\circ \\ &= \sqrt{3}\, V_{ph}\, I_{ph}\, 2\cos\phi \frac{\sqrt{3}}{2} \\ &= 3V_{ph}\, I_{ph} \cos\phi \end{aligned}$$

or,

$$W_1 + W_2 = \sqrt{3}\, V_L\, I_L \cos\phi$$

Thus, it is proved that the sum of the two wattmeter readings is equal to the three-phase power.

Now, we will subtract the two wattmeter readings from each other.

$$W_1 - W_2 = \sqrt{3}\, V_{ph}\, I_{ph}\, [\cos(30° - \varphi) - \cos(30° + \varphi))$$

$$= \sqrt{3}\, V_{ph}\, I_{ph}\, 2 \sin\varphi \, \sin 30°$$

or, $$\sqrt{3}\,(W_1 - W_2) = 3\, V_{ph}\, I_{ph} \sin\varphi$$

or, $$\sqrt{3}\,(W_1 - W_2) = \sqrt{3}\, V_L\, I_L \sin\varphi$$

Dividing added and subtracted wattmeter readings :

$$\frac{\sqrt{3}\,(W_1 - W_2)}{W_1 + W_2} = \frac{\sqrt{3}\,V_L\,I_L\,\sin\phi}{\sqrt{3}\,V_L\,I_L\,\cos\phi} = \tan\phi$$

or,

$$\varphi = \tan^{-1}\frac{\sqrt{3}\,(W_1 - W_2)}{W_1 + W_2}$$

Power factor,

$$\cos\varphi = \cos\tan^{-1}\frac{\sqrt{3}\,(W_1 - W_2)}{W_1 + W_2}$$